

CALCULUS I 2016 Spring Midterm Exam	Dept. or School		Year		proctor	
	Student ID		Name			
※ Your answer must be provided with descriptions how to get the answer. 1. The inverse functions below are considered with the traditional ranges respectively. (a) (4 points) Calculate $\tan^{-1}(\tan(\frac{2\pi}{3}))$. (b) (4 points) Calculate $\cos(\sin^{-1}(-\frac{1}{3}))$.						
(c) (4 points) Show that $\cos^{-1}(\tanh x) = \tan^{-1}(\operatorname{csch} x)$. 2. (8 points) Calculate $\lim_{t \rightarrow 0^+} \frac{(1+2t)^t - 1}{(1+3t)^t - 1}$.						

3. Let V be the volume of the solid obtained by rotating about the y -axis the region bounded by

$$y = e^{(x-2)^4}, \quad x=0, \quad x=4, \quad \text{and} \quad y=0.$$

(a) (6 points) Find the integral representing the volume V of the solid. Note that you do not need to evaluate the volume integral.

(b) (6 points) Let $k = \int_0^2 e^{(x-2)^4} dx$. Express the value of the volume integral in (a) in terms of k .

4. (a) (6 points) Show that

$$0 \leq \int_5^{10} \frac{x^2}{x^4 + x^2 + 1} dx \leq 0.1$$

by comparing the integrand to a simpler function.

(b) (6 points) Find the average value of the function $f(t) = \frac{\cos \sqrt{t}}{\sqrt{t}}$ on the interval $[0, \pi/2]$.

7. (8 points) Evaluate the definite integral:

$$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx.$$

8. (8 points) Evaluate the indefinite integral:

$$\int \frac{dx}{1+e^x+e^{2x}}.$$